



› FOR SOCIETY

SOUND SOURCE LOCALIZATION

PROJECT DEBRIEFING

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VERSION HISTORY

Version	Date	Author	Changes
1.0	12-04-2024	Tom Verlinde, Yassine Tama, Casey Stijlaart, Teun Spithoven	Initial version

INTRODUCTION

This document was converted from MD to Typst. The original text is present on the team wiki

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PROJECT OVERVIEW

CLIENT

The client for this project is Fontys Dienst Huisvestiging & Facilitaire Zaken, in collaboration with Jan Dobbelsteen.

MOTIVATION

In pursuit of an optimized work environment, addressing noise interference emerges as a critical concern. Within the dynamic setting of Fontys Open ICT Lab (OIL), noise disruptions pose significant challenges to productivity and comfort. Recognizing the importance of creating conducive workspaces, mitigating noise-related issues stands as a primary motivation for this project.

TARGET AUDIENCE

The project primarily caters to individuals working within the Fontys OIL environment. As a important working area for various activities, such as collaborative projects and academic pursuits, ensuring a calm and focused work atmosphere is essential for all occupants.

TRL (TECHNOLOGY READINESS LEVEL)

The project currently stands at TRL 3: Experimental proof of concept. This level signifies the initial stages of development, where fundamental principles are being tested and validated in a controlled environment. As the project progresses, it aims to advance towards higher TRL levels through iterative refinement and testing.

PROJECT ASSIGNMENT

CONTEXT

Fontys Dienst Huisvesting & Facilitaire Zaken is a part of the Fontys University of Applied Science, which takes it upon themselves to set the general way of working within the building. As they want to provide an optimized working environment for each individual on campus, they urge to improve the current situation.

In pursuit of an optimized work environment, addressing noise interference emerges as a critical concern. Within the dynamic setting of Fontys Open ICT Lab (OIL), noise disruptions pose significant challenges to productivity and comfort. Recognizing the importance of creating conducive workspaces, mitigating noise-related issues stands as a primary motivation for this project.

The previous group has already made a system that combines embedded software and web applications that together create an application that displays a heatmap of where the sound is coming from. In its current state the solution is only able to calculate the location of the loudest measured sound source.

GOAL

The project goal is to establish an improved version of the current prototype. Besides the improvements, it is of great necessity to develop the prototype even further, by adding new features. The feature which will be added throughout the project is the ability to locate multiple sound sources at the same time.

ASSIGNMENT

The assignment encompasses several key aspects:

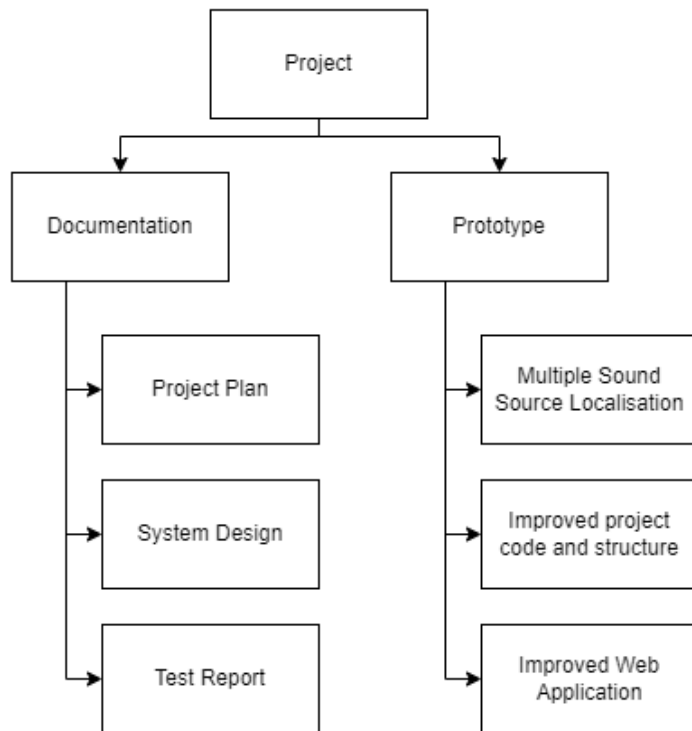
- PCB : A new iteration of the PCB, on which the previous discovered flaws are helped.
- Code Improvements : The current project code status will be improved, to enhance the readability and maintainability, focussing on the SOLID principles.
- Sound Source Localisation : Establish a connection with the system, which is able to localise more than one sound source.
- Web Application : Update the current web application to ensure improved readability and user friendliness.
- Testing Environment : To ensure good code, testing is added to the existing and newly developed code is added.

SCOPE AND PRECONDITIONS

The project includes:	The project does not include:
Test Report, to verify the behaviour of the system.	the Dutch version of documents.
System Design Document.	A fully finished working product.
Firmware in the given programming language.	
A working prototype for multiple sound source localisation.	

Preconditions – during the project multiple technologies will be used to get to the final deliverable. These technologies include Docker, Visual Studio Code, MatLab, and Gitlab. Besides these technologies, other ones will be used as well to develop the firmware, C++, web application, and the necessary hardware.

FINISHED PRODUCTS



RESEARCH QUESTIONS

To get to the final deliverable, multiple aspects first will need to be researched. Each aspect serves a different purpose on how the system should operate. To research the necessary aspects, research questions are created to help guide the research process. There are multiple main questions and multiple sub-questions, the questions are divided regarding specifying how the system should work, which hardware will be used and how they are used.

- What is the current state of the sound source localization?
- Which documentation is there, and which must be worked on to provide a good base?
- How can the current project be used to add the possibility of localising multiple sound sources?
 - ❑ Which hardware is necessary to allow for multiple sound source localisation?
- Which algorithm will be the most suitable for multiple sound source localisation?
- What are the current pain points of the web application?
- Which improvements must be made to the PCB?

RESEARCH METHODS

Each research question requires a different approach for how it will be solved. Therefore, each question will be differentiated and the applicable research method is discussed.

Besides the specific research methods per research questions, some research methods apply to non-specific research questions. Meaning that these research methods are used throughout the whole duration of the project. Such ways are e.g. reviewing and testing (e.g. Unit). As it is of great importance the code and general project is up to the standards and aligns with the vision of the project client.

Research Question	Strategy	Technique
What is the current state of the sound source localization?	Library, Field	Best good and bad practises, Document Analysis, problem analysis.
How can the current project be used to allow add the possibility for localising multiple sound sources?	Library, Workshop	Literature Study, Prototyping.
Which hardware is necessary to allow for multiple sound source localisation?	Library, Workshop	Literature Study, Prototyping
Which algorithm will be the most suitable for multiple sound source localisation?	Library, Workshop	Literature Study, Prototyping.
What are the current pain points of the web application?	Field, Showroom	Survey, Product Review
Which improvements must be made to the PCB?	Field, Lab, Library	Problem Analysis, System Test, Literature Study

RESPONSIBILITIES

Name	Role	Responsibility
Casey	Team Lead	Ensures that all the team members are doing what must be done and that the project continues to go into the correct direction.
Tom	Communication Master	Ensures good communication with the stakeholders and overall team members. Takes care of making any appointments and informing of the necessary people.
Yassine	Scrum Master, Note Taker	Organizes the meetings and standups, and ensures the git scrum board is up to date. Besides that, keeps track of an important notes.
Teun	Git Master	Takes care of Git. Ensuring it is always up-to-date and clean. Besides this, also ensures the necessary environments in git are set up for the project.

APPROACH AND PLANNING

APPROACH

Throughout the project various aspects must be taken into account to ensure everyone is on the same page. Therefore there is made use of a collaboration framework, and its sub parts.

SCRUM

As mentioned, a collaboration framework is used. The chosen framework is the scrum. Scrum is an agile team collaboration framework, which is commonly used within software projects. For the purpose of our project, there is chosen to work with sprints of 2 weeks. It is worth mentioning, that our version of scrum differs from the usual as this is a university project with an end instead of a continuous project.

STAND UPS

Stand ups will be held at the beginning of the day. During the stand up each team member discusses what they have done the previous day, how far they have come with a task, what they going to do today, and if they require any additional help.

The stand-up also serves as a team meeting, where different discussion can be held and decisions made.

RETROSPECTIVES

The sprint retrospective occurs after every sprint. During the retrospective meeting the team will look back at the previous sprint and look at the things which went bad, good, what can be improved, and which next steps are high priority. These become action points on which the team needs to improve upon. A good retrospective also looks if the previous action points have been taken seriously or if they have been ignored.

TEST APPROACH

As it is of high importance the software implementation works flawlessly, testing is implemented. This entails that during the project tests will be written while programming. Besides the software testing, there is used of different types of testing as well. A good example is integration testing, this to ensure that the various system parts work together accordingly.

BREAKDOWN

Down below is initial breakdown of the project. As previously described, the scrum methodology is used, but in an altered version. This as the project breakdown for our project purpose differs from the typical project breakdown for scrum. The breakdown which is used for the purpose of our project is the waterfall method. This as stated before, the general project isn't continuous and has somewhat of a time frame.

ANALYSIS & START-UP

The first phase focusses on analysis and the general start-up. It will cover the first sprint of the project, as in this phase an analysis of the work done by the previous group will be discussed and documented. The outcome of such an analysis results in a clear problem definition, providing us a clear starting point. Besides the overall analysis, the first stage of defining the project is started. Meaning that a first version of the project debriefing is written, as well as the group rules and agreements.

IMPROVEMENTS

During the next phase, improvements to the starting project state is made. The improvements needed will be concluded through the analysis document and project debriefing written in the previous phase. One example of an improvement is the PCB design the previous group designed.

RESEARCH

During this phase additional research towards the end goal is conducted. To define the necessary research, it must be discussed further before providing a concrete subject.

DESIGN

This phase of the project will focus on designing the necessary products, such as the system design, structure, pcb, etc. As these are vital aspects of how the project works. Besides creating design of the current state, new designs, or improved designs, will be made as well.

REALISATION

During this phase the actual code for the project will be written and validated. And as previously described, most of the unit tests will be written and performed during phase as well.

TESTING

In the second last phase, the focus is shifted to a different form of testing. Such tests which cannot be performed with unit tests, especially as the project is focussed on embedded systems, which works with microphones. An example of such tests are integration testing.

EVALUATION

In this last phase the project should be finalized and ready to be presented. Next to this, an evaluation will be made with eventual improvements and/or additions, and an document informing on the state of the project at that point.

TIMELINE

Time	Phase	Tasks
Sprint 0	Analysis & start-up	• Analysis document • Project debriefing
Sprint 1	Improvements, Research	• Implementing possible improvements • Research documentation • Proof of concepts
Sprint 2	Research	• Research documentation • Proof of concepts
sprint 3	Design, Realisation	• Design documentation • Implementation and unit testing
Sprint 4	Realisation, Testing	• implementation and unit testing- Testing
Sprint 5	Testing, Evaluation	• testing • Final realisation • Evaluating • Hand-over documentation

TEAM MEMBERS

Our team consists of the following members:

- Verlinde, Tom T.S.
- Spithoven, Teun T.
- Stijlaart, Casey C.
- Tama, Yassine Y.

We each have our dedicated role within the team on top of the development responsibility.

Name	Role	Responsibility
Casey	Team Lead	Ensures that all the team members are doing what must be done and that the project continues to go into the correct direction.
Tom	Communication Master	Ensures good communication with the stakeholders and overall team members. Takes care of making any appointments and informing of the necessary people.
Yassine	Scrum Master, Note Taker	Organizes the meetings and standups, and ensures the git scrum board is up to date. Besides that, keeps track of an important notes.
Teun	Git Master	Takes care of Git. Ensuring it is always up-to-date and clean. Besides this, also ensures the necessary environments in git are set up for the project.

COMMUNICATION

As a team, we will communicate through Teams or the Whatsapp group. Communication done in person will be written down in either Teams or that Whatsapp group. This way all (official) communication will be written down. Not all in-person communication has to be written down, but agreements or promises should be.

TEST ENVIRONMENT

Throughout the project, multiple manners are used to test and validate the created software and product. This project's three most important testing methods are 'integration testing', 'unit testing' and 'static code analysis'.

INTEGRATION TESTING

Integration tests will be made to test solutions that combine multiple applications.

UNIT TESTING

Unit tests validate each component of code on the lowest level. They validate individual functions and methods. These tests ensure that a function or method behaves as expected.

STATIC CODE ANALYSIS

Static code analysis is done to ensure that code is written in the same style as the existing code. This improves readability and makes it easier for team members to understand code from other team members.

CONFIGURATION MANAGEMENT

To ensure our source code is properly managed we use version control. Specifically GitLab. This means that we aim to have all files present on GitLab, either in the wiki or the main repository.

For task management, we use GitLab's integrated issue tracking in combination with the issue board. We formulate the issues in the form of user stories.

For GitLab runner management we use Docker to ensure the runners contain the exact programs we want on the exact versions we want.

To ensure safety and no interference between development, and code in general, a member creates a branch for the corresponding issue and will work on this branch for the duration of the issue.

Once a group member has finished an issue, the branch may be requested to be merged. This means the branch is up for review and once approved, will be merged to the main branch. Once the branch has been reviewed, approved by two members and all jobs have passed. the merging is executed and the source branch is deleted.

FINANCE AND RISKS

Effective management of finances and anticipation of potential risks are essential components of any project's success. In this section, these aspects are outlined to ensure a clear overview. By understanding the financial resources and proactively identifying and mitigating risks, it is aimed to ensure the smooth execution and achievement of project objectives.

FINANCE

Fontys provides essential support for hardware requirements through its ISSD and Create Lab facilities. Additionally, access to multiple 3D printers is available for any necessary 3D printed components. In the event that specific components are unavailable, discussions with the client can explore alternative sourcing options.

RISKS AND FALLBACK ACTIVITIES

Throughout the duration of the project, several risks may arise, potentially jeopardizing project success. It is crucial to identify these risks and establish contingency plans to mitigate their impact.

Risk	Likelihood	Impact	Preventive Measures	Fallback Activities
Change in scope	Medium	High	Maintain close communication with the project lead to monitor project dynamics and anticipate changes.	Discuss the feasibility of scope changes and evaluate the necessity of altered requirements.
Drop out or absence of team member	Low	Medium	Ensure transparent communication regarding team member availability. If prolonged absences occur, take immediate action to address the situation.	Delegate tasks to other team members and adjust work distribution accordingly.
Hardware issues	Low	High	Plan procurement activities in advance and promptly address any hardware-related issues with the appropriate personnel.	Communicate hardware problems to the team and devise temporary solutions while awaiting resolution. Utilize downtime for documentation tasks.
Miscommunication	Medium	Medium	Attend regular meetings to update team members on progress and plans. Address discrepancies promptly to prevent misunderstandings.	Utilize various communication channels and conduct additional meetings if necessary to clarify misunderstandings. Utilize version control systems like GitLab to track changes and revert any unintended alterations.
Lack of alignment in hardware/software standards	Low	Medium	Conduct regular code reviews to ensure compliance with established standards.	Discuss the situation within the team and find possible solutions on how to make it up to their standards.